

Ousmane Nikiema

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SUMMARY

Electrical Engineering Professional (3.90 GPA) with extensive internship experience in embedded systems, firmware, and PCBA design. Proven track record of developing solutions from concept to prototype, including Zephyr RTOS, BLE protocols, and edge AI. Skilled in C, Python, and VHDL with a focus on power efficiency and real-time performance.

EDUCATION

B.S. Electrical Engineering | Milwaukee School of Engineering | GPA: 3.90 | May 2026

TECHNICAL SKILLS

Programming and Software Development: Embedded systems, bare metal C, C++, Java, Python, VHDL, Quartus Prime, RTOS, PLC programming, IoT, Git, Azure DevOps, CMake, debugging, YOLO

Hardware Circuit Design, and Simulation: LTSpice, Multisim, AutoCAD, Oscilloscope, Soldering, Circuit Design, PCB Design, FPGA, Altium Designer, Colab, ARM Cortex-M

INTERNSHIP EXPERIENCE

Electrical Engineer Intern | Enerpac | May 25' – Now | Milwaukee, WI

- Developed a responsive Zephyr RTOS and LVGL embedded UI with an FSM architecture, integrating multilingual support for 4 languages and custom Chinese font rendering.
- Led hardware bring-up and component selection, delivering prototypes for product integration.
- Engineered a BLE pump emulator on Raspberry Pi using BlueZ stack to facilitate mobile app integration testing of job configuration, monitoring and report feature.
- Contributed to DFMEA by identifying subsystem failure modes and mitigation strategies in design reviews.

Electrical Engineer Intern | Enerpac | Jun 24' – Aug 24' | Menomonee Falls, WI

- Developed C-based firmware using Zephyr RTOS to integrate temperature, current, and pressure sensors via SPI, I2C, and ADC, delivering a fully functional prototype for real-time performance monitoring.
- Architected a custom BLE GATT profile with 128-bit UUIDs, optimizing data serialization to ensure high-fidelity telemetry and 98% sensor data accuracy during functional testing.

ENGINEERING PROJECT EXPERIENCE

Senior Design Project: Underwater Plastic Debris Detection System

- Architected an underwater detection device within a 4-person team, leading the system design from initial requirements through final hardware-software integration.
- Trained and optimized a custom YOLO-based model for real-time edge deployment on the AI processor.
- Developed C firmware with a low-power hardware interrupt to wake the AI unit upon acoustic detection.

Embedded Systems Project – Autonomous Robot (SUMOBOT)

- Integrated motors, sensors, and power system with STM32U575 via custom wiring and 5V regulator.
- Programmed C-based PWM signals to drive motor direction and torque, optimizing speed and frequency.
- Implemented line detection (IR transmitters and receivers) to keep robot within arena boundaries.
- Developed control logic and tracking mechanism using IR sensors to detect and engage opponents.

CO-CURRICULAR INVOLVEMENT

Member | MSOE National Society of Black Engineers (NSBE) | Sept 23' – Present | 2 hrs per wk

Member | Institute of Electrical and Electronic Engineer club (IEEE) | Sept 24' – present | 2 hrs per wk

Member | Madison College STEM Club | September 2022 – May 2023 | 3hrs per wk

ADDITIONAL WORK HISTORY

Undergraduate Research Assistant | MSOE – Fluid Power Institute | Jan 25' – May 25'

Electrical Engineer Intern | Internship On Demand – Bemis Manufacturing | Jun 22' – Aug 22'

Tutor | MSOE – Raider Center for Academic Success | Jan 24' – Dec 24' | 6hrs per wk

Tutor | Madison College Student Achievement Center | Nov 22' – August 23' | 8 hrs per wk

Club Development Coordinator | Madison College Student Life – ELT | Jan 22' – July 23' | 12 hrs per wk